BEU CONNECT

Bihar Engineering University, Patna

Special Examination - 2023

Time: 03 Hours

Code: 100403 ***Subject: Digital Electronics*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Answer any seven question only: **[14]**

(a) Define Dynamic RAM.

(b) The resolution of a 10-bit AD converter for an input range of 10 V is approximately _____

(c) The parameter through which 16 distinct values can be represented is known as _____

(d) Define Multiplexing.

(e) Write the significance of Truth table.

(f) Write the differences between Combinational and Sequential circuit.

(g) A hexadecimal odometer displays F52F. The next reading will be _____

(h) The basic property of Sequential circuit is _____

(i) How many entries will be in the truth table of a 4-input NAND gate?

(j) What is Dual Slope ADC?

Q.2 Solve this question : **[14]**

What are weighted, non-weighted, cyclic and self-complementary codes? Explain each with examples.

Q.3 Solve this question : **[14]**

Design a BCD to 7-segment display decoder circuit using logic gates.

Q.4 Solve this question : **[14]**

Design a 3-bit parallel comparator A/D convertor that provides output in 2's complement format.

Q.5 Solve both questions : **[14]**

(a) Design full adder using 8:1 mux.

(b) Design full adder using 4:1 mux.

Q.6 Solve this question : **[14]**

What is binary shift register? Write down their application.

Q.7 Solve this question : **[14]**

Minimize the following expression using k-map
$f(P,Q,R,S) = \Sigma m(0,1,4,5,7,12,13)$

Q.8 Solve this question : **[14]**

Describe the procedure to design Mod-6 counter.

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store

BEU CONNECT

Q.9 Write short notes on any two of the following: **[14]**

(i) RAM

(ii) ROM

(iii) CMOS logic

(iv) Operation of TTL logic circuit working as NAND Gate

BEU CONNECT



BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified

PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students

GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store

